

Achraf Triki

 Achraf triki
 +216 20656061
 achraf.triki@enetcom.u-sfax.tn
 17 Street Jalta El Bosten, Sfax



Profile

Currently a second-year Industrial Computer Engineering student at ENET'Com, I am highly interested in digital technologies and intelligent systems. My main area of interest is the development of solutions that combine both hardware and software, particularly in the automotive and industrial sectors, with a strong focus on the Internet of Things (IoT), Computer Vision, and Embedded Linux systems.

Projects

- **Real-time finger detection system with Raspberry Pi 4 and ESP32**
Personal Project – IN PROGRESS 2025-2026
 - Real-time finger number detection via camera (Pcamera).
 - Control of LEDs via GPIO ESP32 according to the number of fingers detected..
 - UART communication between Raspberry Pi 4 and ESP32.**Technologies:** Raspberry Pi 4, ESP32, Python, C++/Arduino, MediaPipe, OpenCV, PySerial, UART, GPIO, Linux(Debian)

- **AI-Powered Dermatological Disease Detection System**
Summer Internship – Centre for Digital Research of Sfax (CRNS) 2024-2025
 - Designed and developed an AI-based dermatological disease detection system using YOLOv8 for automatic lesion detection and classification from medical images.
 - Collected, cleaned, annotated, and augmented a dermatological dataset using Roboflow to improve model robustness and generalization.
 - Trained and evaluated the YOLOv8 model using computer vision and deep learning techniques for multiple skin conditions.
 - Developed an interactive web application using Streamlit and Flask for real-time image analysis and visualization of detection results.**Technologies:** Python, YOLOv8, Roboflow, Streamlit, Flask, OpenCV, Computer Vision, Deep Learning

- **Real-Time Object Detection System**
Personal Project 2025-2026
 - Designed and deployed a real-time object detection system on Raspberry Pi 4 using YOLO (YOLOv8/YOLO11), optimized for ARM architecture through the NCNN framework.**Technologies:** Python, YOLO (Ultralytics), NCNN, OpenCV, Picamera2, Raspberry Pi 4, Linux (Debian)

- **Web Application Deployment on a Virtualized Infrastructure**
Personal Project 2025-2026
 - Deployed two virtual machines (Nginx/Flask + PostgreSQL) with a private network, configured UFW firewall rules, and automated backups.
 - Migrated the application to Docker to improve deployment speed and scalability.
 - Implemented a monitoring system using Prometheus for real-time metrics collection and Grafana for performance visualization (CPU, RAM, Disk).**Technologies:** VirtualBox, Nginx, Flask (Python), PostgreSQL, Docker, Grafana, Prometheus, UFW, Bootstrap, Linux (Ubuntu), Networking

Skills

- **Programming Languages:** Arduino IDE (C/C++), VB.NET, Java, Python
- **Generative AI Technologies:** Deep Learning, LLMs
- **Embedded Systems:** ESP32, Raspberry Pi, STM32
- **Databases:** PostgreSQL, MySQL
- **Operating Systems:** Windows, Linux (Ubuntu, Debian)
- **Automation:** Gonenberg, n8n
- **Monitoring:** Grafana, Prometheus

Education

- **Engineering Degree in Industrial Computer Engineering — ENET’Com** 2024-2027
- **Preparatory Studies for Engineering Schools — IPEIS** 2022-2024

Languages

- **French :** Advanced
- **English :** Advanced
- **Arabic :** Native